CHE 565 -- Homework 3

Soki Sem

April 18, 2026

Problem 1

Steepest Descent

Function :

$$f(x_1, x_2) = 0.5x_1^2 + 1.0x_1x_2 - x_1 + 1.0x_2^2 - x_2$$

At x0 =[1, 1], f(x0) = 0.5

Gradient of f:

$$\nabla f = \left[\begin{pmatrix} 1.0x_1 + 1.0x_2 - 1.0 \\ 1.0x_1 + 2.0x_2 - 1 \end{pmatrix}\right]$$

At x0 =[1, 1], Df(x0) = 1.00, 2.00

$$\begin{bmatrix} g_1(\alpha) \\ g_2(\alpha) \end{bmatrix} = \begin{bmatrix} 1 - 1.0\alpha \\ 1 - 2.0\alpha \end{bmatrix}$$

Function phi_k:

$$\phi(\alpha) = 2.0\alpha^4 - 6.0\alpha^3 + 8.5\alpha^2 - 5.0\alpha + 0.5$$

Derivative of phi_k:

$$\frac{d}{d\alpha}\phi(\alpha) = 8.0\alpha^3 - 18.0\alpha^2 + 17.0\alpha - 5.0$$
$$0 = 8.0\alpha^3 - 18.0\alpha^2 + 17.0\alpha - 5.0$$

Optimal alpha: 0.50

Next iterate x1: 0.50, x2: 0.00

Function value at next iterate: -0.38

Problem 2

Conjugate Gradient Method

$$\begin{bmatrix} s_1(\alpha) \\ s_2(\alpha) \end{bmatrix} = \begin{bmatrix} -0.5 \\ -1.5 \end{bmatrix}$$